DEPARTMENT OF THE ARMY
**UNITED STATES MILITARY ACADEMY**
West Point, New York 10996

MADN-MTH 2 November 2026

MEMORANDUM FOR DXDT "Velocity" Scholarship Committee, United States Military Academy, West Point, NY 10996

SUBJECT: Application for AY 2027 DXDT "Velocity" Scholarship

1. Long-term Professional Goals. I intend to serve as an Army officer who turns operational problems into rigorous models and fielded software. I plan to commission as an Infantry officer and lead Soldiers first, because I believe many crucial problems start within formations. As an Operations Research major with a minor in artificial intelligence and a finance specialization, I then want to build toward service as an Operations Research/Systems Analyst (FA49) or work with AI through the Army AI Integration Center (AI2C) or the GenAI Task Force. Additionally, I plan to earn a graduate degree, such as an MBA or another qualifying degree, and return to West Point to teach in the Department of Mathematical Sciences, bringing current industry practice into the classroom.

2. Purpose. Mr. Philippone has described data as the ammunition of the future and has framed a commander's hardest choices as constrained optimization, which means finding the best decision across competing tradeoffs fast enough to matter. This past June, on the Sea to Shining Sea Domestic AIAD, a summer program separate from the fellowship I hold during the academic year, I saw that idea from several angles. Palantir's team in Palo Alto walked us through Project Maven and how it gives commanders at every echelon a common picture to decide from, and the 4th Infantry Division showed us how a formation employs those systems and feeds lessons back to the developers. At Stanford's Gordian Knot Center, Col. (Ret.) Joseph Felter and Jack Faherty described how student startups get new technology into the hands of Army units, and why that is so difficult. At Meta, we met with LTC Andrew Bosworth, Meta's chief technology officer and an Army Reserve officer in Detachment 201, which brings technology executives into the Army.

At my second AIAD, with the Army's Transformation Decision Analysis Center (TDAC), I redesigned a reinforcement-learning model for contested resupply. I am pursuing the DXDT "Velocity" Scholarship for two reasons. First, its funding would take my research out of the classroom to defense conferences where I can present it, a hackathon where I can compete, and an AIAD where I can learn how tools like it are tested before they reach a commander. Second, as a Velocity Scholar I would seek to work alongside industry partners such as Palantir, whether through an AIAD, a hackathon, or a joint research project, and bring that practice back to my research and my future formation.

3. Near-term Research Goals. My interest in contested logistics began at TDAC, where I redesigned the reward function for a reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's sustainment simulation. I built diagnostic tools that

exposed why a high-scoring agent was starving the priority unit, and the retrained agent more than tripled the share of requests filled. That work showed me how a supply system can look successful while failing the unit that needs it most, and it left me asking the opposite question of where an adversary would strike to break one.

That question became my current project with my advisor, LTC Matthew Webb, whose doctoral research was in transportation and logistics. My project asks where an adversary with a fixed effort budget should act to cut Taiwan’s crude oil supply. Additionally, it tests whether LTC Webb’s reinforcement-learning attacker, trained on Ukraine’s rail network, still finds strong strategies on sea lanes. I built a 57-node, 79-edge model of Taiwan’s crude import routes from open sources, with sourced capacities and seven costed adversary actions, and verified it against an independent max-flow calculation. I score each plan by the share of Taiwan’s crude demand that still arrives after the adversary spends its budget, averaged over whether each action succeeds and with Taiwan rerouting supply, so a lower score is a better plan for the adversary.

My near-term goal is to finish this study. Over the coming year I will (1) test whether the trained agent beats random plans, which leave Taiwan about 90% of its demand, and a greedy baseline, which leaves 70%, and comes close to an exact solver while answering in seconds rather than hours, (2) run six wargaming scenarios, from the base case to lower and higher action costs, higher and lower demand, and a closed Strait of Hormuz, to see how the adversary’s best plan changes, and (3) report whether a rail-trained attacker transfers to a maritime network. I will present the results at Projects Day and, if accepted, at SIEDS. In my remaining years at West Point, I want to extend the model from attacking a network to defending one, deciding where added capacity or protection keeps supply flowing, and develop that work into a senior thesis.

4. Funding Plan and Budget.

a. Funding Plan. I plan to attend one AIAD and one conference and to complete three online courses. I list the AIAD and conference options in priority order and will attend only one of each, so each group carries a single estimate, $6,700 for the AIAD and $2,000 for the conference.

   AIAD, Primary. In summer 2027, I will spend $6,700 on housing, food, and transportation for a three-week AIAD with Palantir or Shield AI, the AIADs for which Velocity Scholars receive priority fill. Both companies build the systems that turn models like mine into decisions, Palantir through the Maven Smart System and Shield AI through its autonomy software. Working inside one would show me how and why industry is outpacing the Department of Defense in fielding AI.

   AIAD, Alternate 1. If neither placement is available, I will do a three-week AIAD from 11–31 July 2027 at MIT Lincoln Laboratory in Lexington, MA, a Department of Defense research center that builds and tests AI and decision-support tools for the military. At TDAC, the agent I worked on scored well while starving the priority unit, which is a testing and evaluation failure. Lincoln Laboratory is where I would learn how engineers catch failures like that before tools reach Soldiers.

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AIAD, Alternate 2. My second alternate is a research AIAD at Duke University’s Department of Electrical and Computer Engineering, where LTC Patrick Kuiper earned his doctorate. It ranks third because it is academic research rather than work inside the defense-technology community, but it would still deepen the machine-learning skills my project depends on.

Conference, Primary. On 24–25 May 2027, I will spend $2,000 on train travel, lodging, and meals to attend the SCSP AI+ Expo in Washington, DC; admission is free. It ranks first because it is one of the largest gatherings of defense-technology companies and government AI leaders in the country, with more than 18,000 attendees. If SCSP runs its National Security Technology Hackathon again, I will enter with a team of cadets.

Conference, Alternate 1. In early May 2027, I will spend $2,000 on travel, lodging, meals, and registration to present my Taiwan interdiction paper at the IEEE Systems and Information Engineering Design Symposium (SIEDS) at the University of Virginia, if accepted. It ranks second because acceptance depends on my paper, but presenting would give the project a peer-reviewed, citable publication in IEEE Xplore.

Conference, Alternate 2. From 16–18 March 2027, I will spend $2,000 on travel, lodging, and meals to attend the AUSA Global Force Symposium in Huntsville, AL; registration is free for military attendees. It is the Army’s annual forum on modernization, where senior leaders and industry discuss the technology, acquisition, and concepts that will shape the future force. It ranks third because I would attend to learn rather than to present or compete.

Online Courses. During 2027, I will spend $797 on verified certificates for three courses in the MITx Supply Chain Management MicroMasters program, offered by the MIT Center for Transportation & Logistics. The courses are Supply Chain Analytics (SC0x, $199), Supply Chain Fundamentals (SC1x, $299), and Supply Chain Design (SC2x, $299). Supply Chain Design covers network design and optimization, the defender’s side of the problem my Taiwan model attacks. The courses also count toward the MicroMasters credential, a pathway to graduate study in supply chain management.

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b. Research Travel. Options are listed in priority order. I will attend one AIAD and one conference, so each group has one estimated cost.

| Purpose of Trip (AIAD, Conference, other) | Destination | City, State | Dates or # of Days | Estimated total Trip costs |
|---|---|---|---|---|
| AIAD, Primary | Palantir or Shield AI | Per placement | Summer 2027 (21 days) | $6,700 |
| AIAD, Alternate 1 | MIT Lincoln Laboratory | Lexington, MA | 11–31 Jul 2027 | |
| AIAD, Alternate 2 | Duke University ECE | Durham, NC | Summer 2027 (21 days) | |
| Conference, Primary | SCSP AI+ Expo and hackathon | Washington, DC | 24–25 May 2027 | $2,000 |
| Conference, Alternate 1 | IEEE SIEDS, UVA | Charlottesville, VA | Early May 2027 | |
| Conference, Alternate 2 | AUSA Global Force Symposium | Huntsville, AL | 16–18 Mar 2027 | |

c. Equipment, Materials, or Supplies. Three MITx course certificates purchased from MIT (learn.mit.edu), SC0x ($199), SC1x ($299), and SC2x ($299), for a total of $797.

d. Total. $9,497 (one AIAD $6,700, one conference $2,000, online courses $797), within the $10,000 limit.

e. Timeline for Expenditures. From November 2026 to February 2027, I will request my AIAD choices in priority order, submit the SIEDS abstract, and register for the conferences. I will attend the funded conference between March and May 2027 and complete the AIAD in summer 2027. I will buy each MITx certificate when that course opens, and all funds will be spent by 31 December 2027, before my May 2029 graduation.

5. The point of contact for this memorandum is CDT Dennis Pezan at (815) 592-6927 or dennis.pezan@westpoint.edu.

Encl
Applicant’s Curriculum Vitae

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